HDOC Day-11

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice: Find the sum of squares of its digits using a while loop. E.g. Number: 123, Sum=14

1.Algorithm

1. Start.
2. Display the menu:
   - 1. Find sum of squares of digits
   - 2. Exit
3. Input the user's choice.
4. Use switch(choice).
5. For **case 1**:
   - Input a positive integer n.
   - Set sum = 0.
   - Repeat while n > 0:
     - Find last digit: digit = n % 10
     - Add its square: sum = sum + digit * digit
     - Remove last digit: n = n / 10
   - Display the sum.
6. For **case 2**, exit the program.
7. For any other choice, display "Invalid choice".
8. Stop.

2. Test case table

Test Case	Input Number	Expected Output	Actual Output	Result
1	123	Sum = 14	Sum = 14	Pass

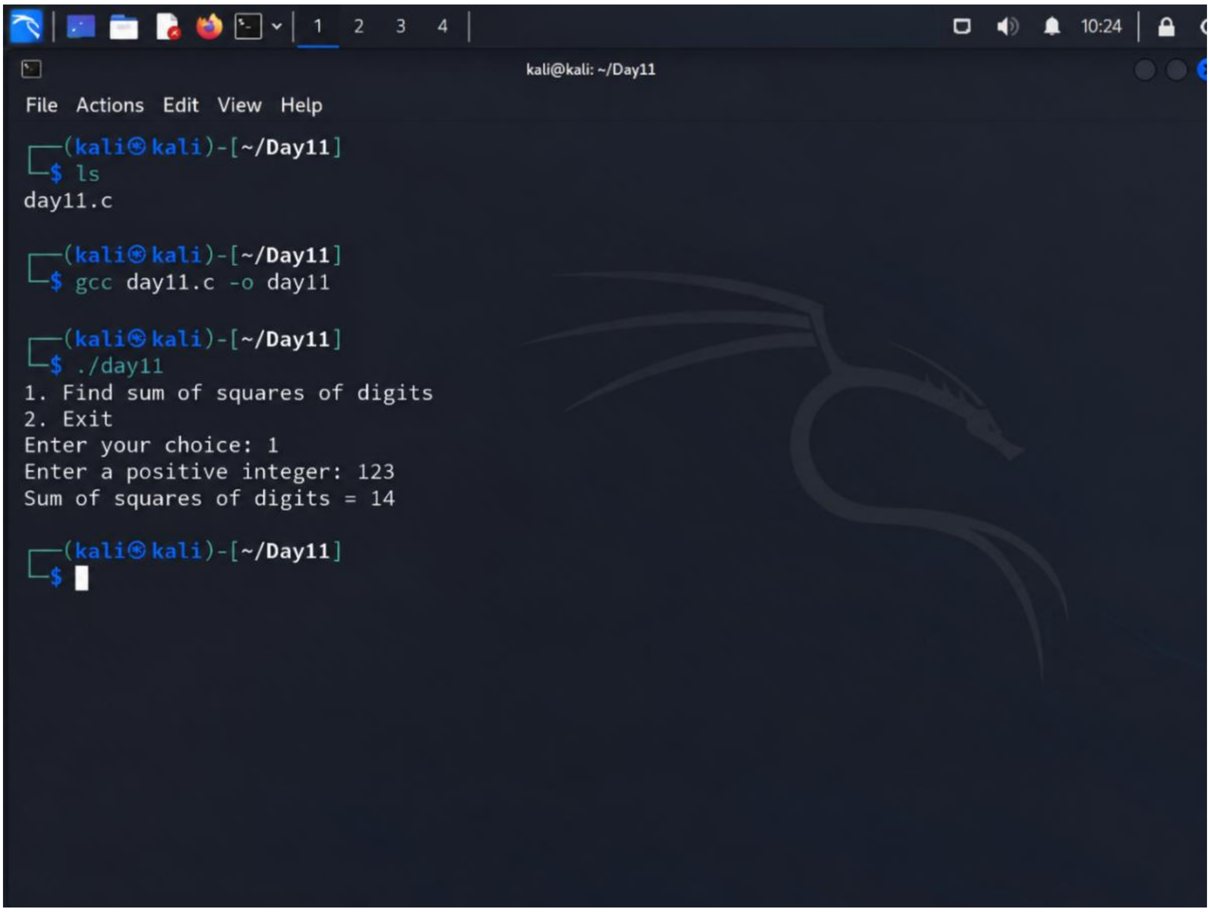
Test Case	Input Number	Expected Output	Actual Output	Result
2	246	Sum = 56	Sum = 56	Pass
3	111	Sum = 3	Sum = 3	Pass
4	505	Sum = 50	Sum = 50	Pass
5	987	Sum = 194	Sum = 194	Pass

3. C Program

```
#include <stdio.h>


int main()

{

  int choice, n, digit, sum = 0;


  printf("1. Find sum of squares of digits\n");

  printf("2. Exit\n");

  printf("Enter your choice: ");

  scanf("%d", &choice);


  switch(choice)

  {

    case 1:

      printf("Enter a positive integer: ");

      scanf("%d", &n);


      while(n > 0)

      {

        digit = n % 10;

        sum = sum + (digit * digit);
```

```
            n = n / 10;
        }

        printf("Sum of squares of digits = %d\n", sum);
        break;

      case 2:
        printf("Exiting program.\n");
        break;

      default:
        printf("Invalid choice.\n");
    }

    return 0;
}
```

4. output



Verification: For 123: $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$, so the program gives the correct result.